Capacitively-Coupled Chopper Instrumentation Amplifier for Linear Hall Effect Sensor

IEEE SSCS Chipathon 2026

Team Chop

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Contents

1 Project Information

Project Name	CCIA for Linear Hall Effect Sensor
Team Members	alang
Review Date	
Review Version	
Track	B

[1]

2 Design Objective

- What problem does the circuit solve
- Target specifications
- Intended use within the Chipathon project

3 System Overview

- Block Diagram
- Description of each block
- Signal Flow
- Power Domains
- Clock Domains

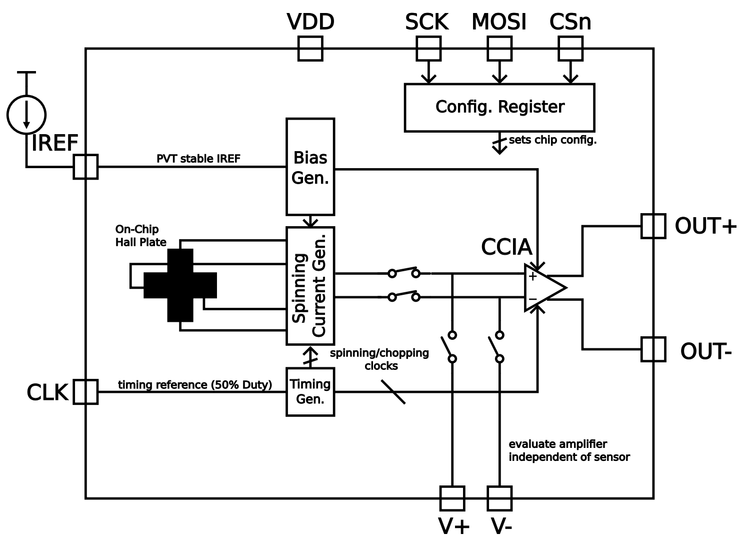

Figure 1: Block diagram of the proposed architecture.

Fig. 1
Table 1

Table 1: Chip pinout description.

Analog Pins		
IREF	Analog	Current reference
OUT+	Analog	Differential output positive terminal
OUT-	Analog	Differential output negative terminal
VTP+	Analog	Positive analog test point (differential probe node)
VTP-	Analog	Negative analog test point (differential probe node)
Digital Interface		
SCK	Digital Input	SPI configuration clock
MOSI	Digital Input	SPI configuration data input
CSn	Digital Input	SPI chip select
CLK	Digital Input	Chopping/spinning clock reference
Supply Pins		
VDD	Power	Supply input
VSS	Power	Ground (shared with other Chipathon blocks)

4 Schematic Summary

4.1 Sheet 1.sch

- Purpose
- Inputs and Outputs
- Important Design Decisions
- Interfaces to other blocks

4.2 Sheet 2.sch

- Purpose
- Inputs and Outputs
- Important Design Decisions
- Interfaces to other blocks

5 Design Assumptions

- Supply Voltages
- Temperature Range
- Process Corners
- Expected Loads
- External Circuitry

6 Verification Status

6.1 Functional Simulations

6.2 Corner Simulations

6.3 Monte Carlo

6.4 Sign-off

- ERC Status
- DRC Status
- LVS Status

7 Design Checklist

7.1 Functionality

7.2 Analog

7.3 Digital

7.4 Mixed-Signal

7.5 Reliability

7.6 Documentation

8 Open Issues

List of open issues with:

- Description
- Impact
- Proposed Solution
- Owner

9 Questions for Reviewers

10 Review Outcome

Appendix

References

[1] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.